

EXERCISE I

In reference to the Stochastic Gradient Descent (SDG) algorithm

I.a) Explain the concept of loss function and provide one example.

I.b) Why do some loss functions use a regularization term?

I.c) Is SGD guaranteed to minimize the loss function? Motivate your answer.

EXERCISE II

Consider a convolutional neural network layer with the following characteristics: 80 input channels (32x32), convolutional filters size 5x5, 160 output channels. Padding with zeros and stride 1.

II.a) Compute the number of MAC operations needed to compute a single output channel by applying convolution on a single input channel.

II.b) Compute the total number of MAC operations needed to compute all the output channels in the layer.

II.c) Compute the number of SUM and number of DIV operations needed to perform average pooling (2x2) on all the output channels.

EXERCISE III

A parallel cluster of RISC-V processors is executing a convolutional layer with 9 input channels and 17 output channels. Input spatial size is 128x128. Inputs, outputs, and weights are quantized to 8 bits. Convolutional filter size is 3x3, no padding and stride 1 is used. RELU non-linearity is applied in place on the output. Assume that on average a RISC-V processor can perform a MAC in 2.5 cycles and a RELU operation in 3 cycles. RELU is performed by a single processor in the cluster, while the convolutions are parallelized by assigning each processor to one or more output channels, aiming at balancing execution time, if possible.

III.a) Compute the L1 minimal shared memory size that is needed to store the complete layer (input and output channels as well as weights) in L1.

III.b) Compute the total number of cycles to process the layer on an 8-core cluster.

III.c) Compute the worst-case percentage of idle time for a core in the cluster in case it's not possible to balance the workload.

